

23.1 Find the speed of light in water ($n_{\text{water}} = 1.3$, $c = 3 \times 10^8 \text{ m/s}$).

23.3 A lamp emits yellow light with wavelength 600nm in air. (a) What is the wavelength of light in water?

23.6 A lamp emits light at a frequency of $5 \times 10^{14} \text{ Hz}$ (a) find the wavelength in the air (b) if the light enters glass with index of refraction 1.5, find the frequency and the wavelength.

23.7 What fraction of the light intensity is reflected when light is normally incident in air on water? What fraction of light intensity is transmitted on water?

23.12 A ray of light is incident on a glass-water surface from glass at an angle of 45° . Find the angle of refraction if the index of refraction of the glass is 1.5.

23.18 What is the critical angle for total internal reflection in diamond? (Assume the diamond is in air and the refractive index of the diamond $n = 2.417$)

23.20 The critical angle for light going from glass to air is found to be 36° . What is the refractive index of the glass?

23.22 Two narrow slits 0.2mm apart are illuminated with red light of wavelength 700nm. At what angles are the five maxima closest to the center of the pattern formed?

23.23 A double-slit apparatus is illuminated with yellow sodium light ($\lambda = 589\text{nm}$). The first maximum fringe is 1cm apart from the central fringe. If the screen 1m away from the double slits, find the slit separation.

23.26 A diffraction grating with 5000 lines per centimeter is illuminated with yellow light of wavelength 589nm. What is the angular position of $m = 1$ line?

23.35 A narrow slit is illuminated with yellow light of wavelength 589nm. If the central diffraction maximum extends from 0° to 40° , how wide is the slit? Make a sketch

23.38 A circular aperture of radius 10^{-5}m is illuminated with light of wavelength 600nm. At what angle is the first diffraction minimum?